Housing First or Treatment First?

A stylized model, and the quantitative model behind it

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1 A stylized model

1.1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder (SUD), and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and fifty four parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This note strips the environment down to help understand the main drivers in the model, but where my three main questions are still well posed.

Households are perfectly forward looking, here and in the quantitative model. They face the true hazard of acquiring a SUD, they know what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, on myopia, or on a taste that changes once the household has used.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs here are a *luxury* good bought at a price: the Stone-Geary form in equation (1) below means the share of income spent on them rises with income, so left to itself the environment concentrates use at the *top*. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, and where the two programmes cost the same to run, never. Housing First always raises welfare when agents are forward looking, as they are here. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension does not raise welfare, because Treatment First only removes an option that a household can take under Housing First as well. However, if a SUD and homelessness generate externalities, such as affecting the earning potential of the children of parents with a SUD, then Treatment First may be preferred: children of a parent with a SUD are themselves two to four times as likely to develop one (McGovern et al., 2023), and moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent (Chetty et al., 2016). The model's work here is to show why Housing First raises welfare whenever there are no externalities (and no distortions from the programmes or from the taxes that pay for them, both of which the full model takes up), and then show how the externality that overturns it works.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains are the main drivers of the results. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

1.2 The environment

1.2.1 Households and preferences

There are two periods and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is β per period, the gross interest rate is $R = 1/\beta$ so that $\beta R = 1$, and ϕ_0 is survival over thirty years. Notation follows the quantitative model throughout.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Households differ in y, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption c, housing services h and drug consumption v,

$$u(c, h, v) \;=\; (1 - \gamma - \theta)\log c \;+\; \gamma \log h \;+\; \theta\big[\log(b + v) - \log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so γ is the housing budget share. Drugs enter only in the first period, since that is the only period in which v is chosen, and housing is chosen each period from a three point grid $\{h_1, h_2, h_3\}$. The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b + v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non users at all, because under plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

1.2.2 Timing

Period 1. Every household is clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability u_0, and a household that is income eligible draws a voucher with probability π_V; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent p_r, which is normalised to one, on whatever unit it picks. It starts with no assets. Using drugs generates the risk of a SUD, increasing in v,

$$\pi(v) \;=\; 1 - e^{-\lambda v}. \tag{2}$$

Because π is increasing in v, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

The transition. The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(d') \;=\; \phi_0\,(1 - \kappa d'), \tag{3}$$

so κ is the proportional reduction in survival caused by the SUD. The mortality penalty applies here, and here only.

Period 2. Terminal, after which the household dies with certainty. A household without the SUD is unemployed with probability u_0 and one with the SUD with the higher probability u_1; if unemployed it receives income support. The voucher is drawn again, subject to the policy rule. The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality, and where the SUD is realised. Mortality is a tax on a continuation value, so the SUD can only cost anything if a future remains when it is carried. That does not require a third period. It requires the SUD to be drawn at a *transition* rather than inside the terminal period, and here it is drawn on the way into period 2, with survival over that transition depending on it, so acquiring it puts the whole of period 2 at risk. Realise it inside the last period instead and death costs nothing because nothing follows it, and the mechanism the model rests on falls silent.

1.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the SUD. Income in a period, given permanent productivity y and employment status e, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e=1, \\ \psi \cdot \min\{y, \bar{y}\} & \text{if } e=0, \end{cases} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of ψ is what links the SUD to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h_k \in \{h_1,\ h_2,\ h_3\}, \qquad h_1 < h_2 < h_3, \tag{5}$$

homelessness (h_1), a small unit (h_2) and a standard unit (h_3). Homelessness is free and delivers less housing services than the market does.[1]

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option k is

$$r_k(\eta) \;=\; p_r\, h_k \big[\kappa_r \eta + (1-\eta)\big] \cdot \mathbf{1}\{k \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in both periods, and π_V, the share of eligible households actually covered, is the policy dial of Section 1.5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of (d', v). **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since policy coverage in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

The condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before v is chosen and conditionality cannot screen anyone out on entry. The order within the period is therefore: the household chooses v, the policy rule is then applied to the voucher it drew, and it chooses housing knowing what rent it faces. So under Treatment First a household that uses would be subject to market rent immediately and again in period 2.

1.2.4 The household's problem

Let a be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$W_2(a,y,e,\eta,d) = \max_k \Big\{ (1-\gamma-\theta)\log c + \gamma \log h_k + \aleph \Big\}, \qquad c = Ra + y^g(y,e) - r_k(\eta), \tag{8}$$

and write $W_2(a,y,d) \;=\; \mathbb{E}\big[W_2(a,y,e,\eta,d) \;\mid\; y, d\big]$ for its expectation over the period 2 employment and voucher draws. In period 1 the household draws (e,η) and then chooses housing, drug use and saving,

$$V_1(y,e,\eta) = \max_{k,\, v\ge 0,\, a'\ge 0} \Big\{ (1-\gamma-\theta)\log c + \gamma \log h_k + \theta[\log(b+v) - \log b] + \aleph + \beta\, \mathbb{E}_{d'} \Big[\phi(d')\, W_2(a', y, d') \Big] \Big\}, \tag{9}$$

with $c = y^g(y,e) - r_k(\eta) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). Survival sits inside the expectation over d' because it depends on which d' is realised.

A household uses if the value at its best positive v exceeds the value at $v = 0$. Households differ only in y and in the draw (e,η), which is the same lottery for everyone, so within a draw the decision defines a set of income intervals on which use occurs, and participation at a given income is the probability of drawing a state in which the household uses.

[1] Two rented sizes is the fewest that lets a household economise on housing rather than lose it. With one, homelessness comes out at 14.40 percent against a target of 0.40, at any h_1.

1.3 Calibration

1.3.1 External calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the discount factor over a thirty year period is $\beta = 0.860$[2] and the gross interest rate is $R = 1/\beta$, the dispersion of log permanent income is $\sigma_y = 0.75$ from Heathcote, Perri and Violante (2010), and survival over a period is $\phi_0 = 0.91$, from the Social Security Administration (SSA) period life table for twenty five to fifty five. For the SUD, the proportional fall in survival is $\kappa = 0.49$, derived by compounding an annual mortality ratio of 8, the midpoint of the range in Hjorthøj et al. (2023), so a clean household's chance of dying before period 2 is 9 percent and a household that acquires a SUD faces 53 percent, and the two unemployment probabilities are measured directly. $u_0 = 0.049$ is the headline civilian unemployment rate from the Bureau of Labor Statistics (BLS) Current Population Survey, series UNRATE as distributed by Federal Reserve Economic Data (FRED), averaged over 2014 to 2024. And $u_1 = 0.1625$ is from the National Survey on Drug Use and Health (NSDUH) 2024 for respondents aged 26 and over.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The two rental sizes are set such that the standard unit costs 0.41 of mean income and the small unit is a quarter of it. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the assisted eligible share in the 2023 Picture of Subsidized Households, published by the Department of Housing and Urban Development (HUD). The weight on drugs is set at $\theta = 0.23$.[3]

1.3.2 Internal calibration

Five parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly. The weight on housing γ is matched to the 26.8 percent of total expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX), housing over expenditure net of drugs. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit b is matched to a drug participation rate of 5.13 percent from NSDUH 2024 for respondents aged 26 and over, housing services on the street h_1 to a homelessness rate of 0.399 percent from the 2023 HUD point in time (PIT) count, converted to households, and the SUD hazard λ to the 33.2 percent of users who acquire a SUD.[4]

1.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: the average user then earns 2.62 times the population average.

Two features of the environment are candidates to reverse it. The *mortality channel* is that the SUD increases the chance of dying, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose

[2]The annual discount rate here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon rather than making a claim about preferences.

[3]A θ large enough to match the eleven percent of income that users spend on drugs makes the drug too valuable to be deterred by a death that arrives thirty years later, and users stop being poor. Setting $\theta = 0.23$ gives a spending share of 6.32 percent and keeps the income gradient, which is the trade this model can make.

[4]The quantitative model matches the level of the disorder rate instead, 3.68 percent, for the reason given with λ in Section 3: the conversion rate and the disorder rate published in the same survey are not consistent with each other, and the level is the better measured of the two. Nothing here turns on which is used, since the two period model has no separate disorder level to reproduce.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor over a thirty year period	0.99 annually, compounded	0.860
R	Gross interest rate over a thirty year period	Set to $1/\beta$	1.162
σ_y	Dispersion of log permanent income	Heathcote et al. (2010)	0.75
ϕ_0	Survival over a period, no SUD	SSA period life table, ages 25–55	0.91
θ	Weight on drug use	Set; see the text	0.23
The SUD			
κ	Fall in survival with a SUD, over a thirty year period	Hjorthøj et al. (2023), age 26+	0.49
u_0	Unemployment probability, no SUD	BLS CPS via FRED, 2014 to 2024	0.0485
u_1	Unemployment probability, with a SUD	NSDUH 2024, age 26+	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Normalisation	1.00
p_v	Price of a unit of drugs	Normalisation	1.00
h_3	Standard rental unit	Set	0.41
h_2	Small rental unit	Set	$0.25 \times h_3$
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD 2023, assisted eligible share	0.27

Note: every parameter here is set from outside, so none has a model counterpart to report against the data. This is the same distinction the Model column of Tables 5 and 6 makes with n.a., which those tables need because they carry externally set and internally calibrated parameters together.

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
γ	Weight on housing services	Housing over non drug spending	26.80%	26.58%	0.289
$\aleph$	Flow value of being alive	Value of a statistical life	80.0	80.0	2.790
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00316
h_1	Housing services when homeless	Homelessness rate	0.40%	0.42%	0.00573
λ	Addiction hazard per unit of use	Share of users who acquire a SUD	33.2%	35.5%	132.6

your home. Figure 1 scales each from nothing to 125 percent of its calibrated value, recalibrating b at every point so participation stays at 5.13 percent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

The result is sharper than the statement that mortality matters more. That channel takes the income ratio down to 0.45 at full strength, and it does so as a threshold rather than gradually: at half of its calibrated value the ratio is still high, at 3.33, and the reversal happens over the last tenth or so of the channel. Past the calibrated point it keeps biting, reaching 0.19 at 125 percent, so the reversal is not a knife edge that the calibration happens to sit on. The unemployment channel does not reverse it at all, leaving users at 3.33 times average income even fully switched on, and at 125 percent it is 3.33, still above one and no closer to turning. Strengthening the resource penalty by a quarter does not begin to do what mortality does. A resource penalty does not produce the concentration of drug use at the bottom.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

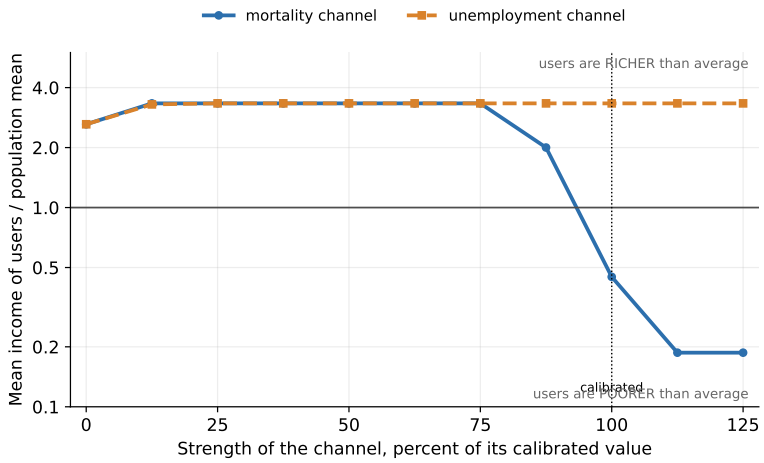


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that participation is 5.13 percent throughout. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

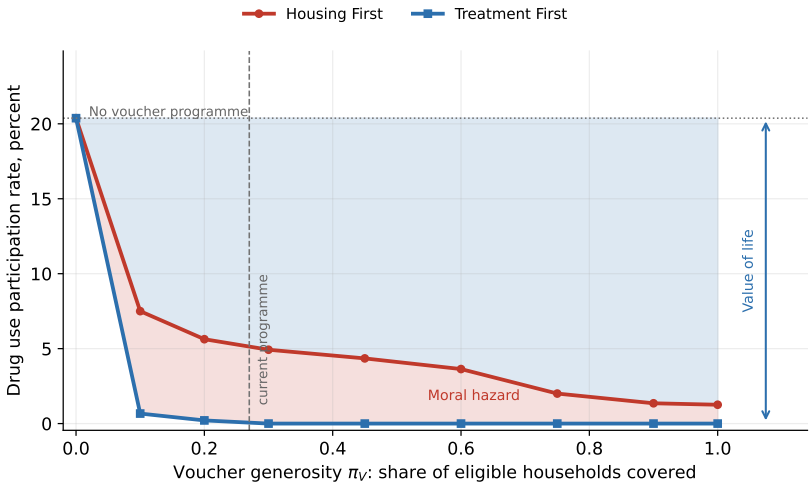


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

1.5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

Figure 2 traces both as generosity rises. With no programme at all, 20.38 percent of households use. Universal Treatment First takes that to 0.00 percent, a value of life force of -20.38 percentage points; universal Housing First leaves it at 1.26 percent, so the moral hazard force is +1.26 points and the net effect of Housing

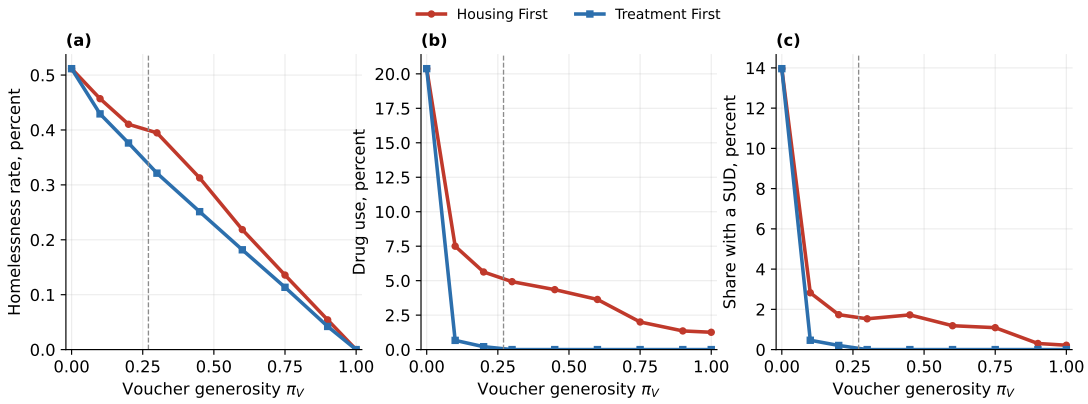


Figure 3: Homelessness, drug use and the SUD under the two policies

First is -19.12. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and SUD outcomes for each programme as its generosity changes, alongside drug use. In panel (a) the two policies end in the same place on homelessness, both driving it from 0.51 percent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the SUD that puts their housing at risk. Panel (b) repeats Figure 2 here for convenience. The larger difference is in panel (c), where at universal cover Housing First leaves 0.217 percent of the population carrying the SUD and Treatment First leaves 0.000 percent. What the moral hazard buys is extra SUD cases among households that are housed either way.

1.6 Welfare

1.6.1 Without externalities, Housing First always wins

Sections 1.4 and 1.5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity π_V, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off under Housing First:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[5]

It follows that the moral hazard is not an inefficiency. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +1.26 points of Section 1.5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 2.7 percent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.50 percent of consumption. Those are exactly the households that use: a household that never uses faces the same voucher rule under both policies

[5]It does, however, take the price of the drug and the cost of the programme as given, both of which this model fixes and the quantitative model does not. There the tax rate that pays for the programme is endogenous, so a policy that costs more raises the tax on everybody, and the drug price moves with the number of users. Either can separate the two policies with no externality present at all, which is why the argument here is stated for this model and not asserted of the other one.

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	1.26%	0.00%
Share with a SUD	0.196%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.3185	0.3185
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		2.7%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.45%
Consumption equivalent loss, whole population		0.012%

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household over both periods, the second discounted and weighted by survival, per unit of average income.

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's SUD rate	Ratio	Loss to the child
0.00	2.24%	1.00	0.0%
0.20	4.04%	1.80	38.7%
0.30 (calibrated)	5.45%	2.43	69.2%
0.40	7.43%	3.31	113.3%
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.012%
Disorder cases prevented			0.196 pp
Damage to the child of one such case			54%
Externality gain from conditionality			0.106%
Gain over loss			8.9

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Losses are consumption equivalents.

and is indifferent between them. The figure is not the 5.13 percent participation rate of Section 1.3 because the programmes compared here are universal, rather than the rationed benchmark that rate is measured at.

A fixed budget does not rescue the welfare case for conditionality either. Treatment First is almost exactly as expensive as Housing First at universal cover, 0.3185 against 0.3184, so matching on cost rather than on generosity changes nothing. Two things offset. Survival into period 2 falls from 0.91 to 0.47 with a SUD, so the extra households Housing First puts into the SUD state die sooner and fewer are alive in period 2 to collect a voucher; against that, Treatment First withdraws the voucher from anyone who used, so it pays for fewer tenants while they live. The expected outlay remains the same. So a case for conditionality has to come from an externality, and, in this model, from nothing else: with the two outlays equal and financing outside the model, there is no distortion left to tell them apart.

1.6.2 An externality on children reverses it

I model an externality that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste, a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. So the specification says that a child who inherits a stronger taste for drugs is *made better off* by inheriting it, and that is the opposite of what these children report. Growing up with household substance abuse is one of the adverse

childhood experiences in Felitti et al. (1998), and it predicts higher rates of depression, worse self reported health and lower life satisfaction in adulthood; the children of parents with a SUD also have lower educational attainment, which is the premise of the transmission literature δ_d is set against (McGovern et al., 2023). A model in which the transmitted trait raises the child's own welfare cannot be describing any of that, whatever it does to the transmission rate. Letting it inherit a greater susceptibility, a higher λ, fails another way, since a child facing a riskier drug optimally uses *less* ex ante, which is at odds with the data: children of parents with a SUD are *more* likely to use, not less.

Income works on both counts, and it works through the channels explored in Section 1.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the evidence, and it stands in for more than income alone: a parent who diverts spending and time to drugs invests less in the child's education and development, and the income penalty is the reduced form of all of that (Chetty, Hendren and Katz, 2016; McGovern et al., 2023). The income gradient does the whole job, so the model transmits a SUD across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{11}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into V_1.[6]

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the SUD 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. The homelessness penalty δ_h is set to 31 percent, directly from Chetty et al. (2016), who find that children who moved to a low poverty neighbourhood before age 13 went on to earn 31 percent more as adults. The two penalties also do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.013 percent of consumption, small because only 2.7 percent of households are affected. Against that, conditionality prevents 0.217 percentage points of SUD cases in the second row, and each of those cases costs one child 54.3 percent of consumption, shown in the third row. The gain in the fourth row is the product of those two, since one prevented case spares one child: 0.217 percentage points times 54.3 percent gives 0.118 percent of consumption, which is 8.9 times the private loss of the first row. That ratio exceeds one for every δ_d in Panel A except $\delta_d = 0$, where by construction there is no externality to price. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

All of this is a crude illustration: the model is two periods long and nobody who faces the sobriety condition fails to meet it, so the impacts of policies can be unreasonably large. It shows that an externality can reverse a welfare ranking that is otherwise airtight, which is all it is for. Which way the comparison actually goes is a question about magnitudes, and that is what the quantitative model of the next section is for.

The quantitative model carries two further channels absent here, and they are different in kind as well as in sign, which is another substantive reason the welfare ranking is a question of magnitudes. The first is not an externality at all but the distortions the programme creates and the distortionary taxes required to pay for it; the second is an externality of the same type as the one on children.

- **Programmes distort the decisions of recipients and are paid for out of distortionary taxes.** A programme lowers the labour supply of the households it does reach, because eligibility is means tested, and the voucher is clawed back at a rate of κ_r, so a recipient who works more loses eligibility or some of the transfer. This, combined with the pure financing of the programmes, means that other households have to pay more taxes, which distorts their labour supply.

[6] In the quantitative model the altruism of parents towards their children is calibrated rather than set to zero, so that the channel where a parent does take some account of what it does to its child is present and only the remainder is external.

- **Homelessness is a disamenity in itself.** Households dislike homelessness around them and not only their own, so the aggregate rate is an argument of everyone's utility. This one is an externality, and it survives any way of paying for the programme. It runs in favour of whichever policy houses more, which is the opposite direction from the tax.

2 The quantitative model

This section describes the model the paper's numbers come from. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below. Notation follows Section 1 wherever the two models share an object: γ and θ are the housing and drug weights in both, b is the same Stone-Geary endowment, $\aleph$ the same flow value of being alive, λ the same SUD hazard, ψ and $\bar{y}$ the same replacement rate and earnings base cap, r the same risk free rate, and β is the discount factor and nothing else.

2.1 Demographics, the state and the choice set

Households enter as adults, live at most J annual periods and retire at J_R, and j indexes age within that span. Survival from j to $j+1$ is $\phi(j, h', d, m, \tau)$ in (24), and it depends on age, on the housing *chosen* this period, on the SUD and on serious mental illness (SMI), and on whether the household is in treatment, since treatment means it is not using, and thus is less likely to die. Newborns replace the dead exactly, so the mass of households is constant. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{12}$$

where a denotes assets, y permanent productivity, e employment status, d the SUD, m SMI and η receipt of a voucher.

Each period the household chooses

$$\left(n,\, h',\, v,\, \tau,\, a'\right) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{13}$$

where n denotes hours, h' the housing unit chosen, of which h_1 is homelessness, v drug use, τ SUD treatment take up, available only when $d = 1$ and requiring $v = 0$ in the period it is taken, and a' next period assets.

2.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\left[\, c^{1-\gamma-\theta}\,(h')^{\gamma}\,(v+b)^{\theta}\,\right]^{1-\sigma}}{1-\sigma} \;+\; \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} \;+\; \aleph \;-\; \tau\bar{\tau}^{u} \;-\; \kappa_x H, \tag{14}$$

Four of these are the stylized model's and carry the same meaning here as in Section 1.2: γ the housing weight, θ the drug weight, so that $1-\gamma-\theta$ is the weight on consumption and the three sum to one; b the Stone-Geary endowment; and $\aleph$ the flow value of being alive. Equation (1) is the logarithmic case of this bundle with hours fixed.

The rest are new, and there are three of them. Leisure makes hours a margin, at weight ω_{ℓ} and curvature σ_{ℓ} over the $1-n$ hours not worked. SUD treatment costs utility, $\bar{\tau}^{u}$ in each period the household takes it up, $\tau = 1$. And homelessness carries a disamenity, κ_x per unit of the aggregate homelessness rate H, borne by every household and priced on the headcount alone, which is what makes homelessness an externality here rather than only a private misfortune; Section 3.1 takes it up. Outside the bundle, σ is the coefficient of relative risk aversion.

2.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\,y & j \geq J_R \quad \text{(social security)} \\ n\,y\,\varepsilon(j)\,\rho_d^{\mathbf{1}\{d=1\}}\rho_m^{\mathbf{1}\{m=1\}} & j < J_R,\ e=1 \quad \text{(employed)} \\ \psi\ \min\{y\,\varepsilon(j),\ \bar{y}\} & j < J_R,\ e=2 \quad \text{(first year of a spell)} \\ \zeta_{DI}\,y\,\varepsilon(j) & j < J_R,\ e=3,\ m=1 \quad \text{(non employed with SMI: disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

taking the cases in the order they appear. Retirement income replaces 40 percent of the productivity state with no health penalty, since social security is not means tested on health. For the employed, $\varepsilon(j)$ is the deterministic age profile of efficiency and ρ_d and ρ_m are productivity penalties applied *conditional on working*. The three employment states are work, $e=1$, the first period of a spell out of work, $e=2$, and non employment thereafter, $e=3$, so the last two cases are both unemployment and the disability case is the income of a household that is out of work and carries SMI. Unemployment insurance (UI) is paid in the first period of a spell only, at the replacement rate ψ on an earnings base capped at average income $\bar{y}$, both as in equation (4). ζ_{DI} is the rate at which disability insurance (DI) replaces the income of a non employed household with SMI.

The two penalties, ρ_d and ρ_m, are in the model to generate a large share of homeless households being employed, consistent with the data. Employed homelessness cannot be produced by unemployment alone: something has to leave a household poor while it is still in a job. Both are calibrated internally: ρ_d on the share of the homeless who are employed, and ρ_m on the earnings of workers with SMI relative to average.

Labour income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate τ_k,

$$y^n = \tau_\ell\,(y^g)^{\xi} \ + \ (1-\tau_k)\,r\,a, \tag{16}$$

with y^n denoting after tax income, where ξ governs progressivity and τ_ℓ the level. Households can receive two transfers, both means tested on gross income as the programmes themselves are. A food stamp floor tops means tested income up to $\underline{c}\,\bar{y}$, with $\underline{c}$ the floor expressed as a share of average income, and the voucher covers rent above a tenant contribution of κ_r of gross earnings, the same κ_r as in Section 1.2.3,

$$F = \max\{0,\ \underline{c}\,\bar{y} - (y^g + ra)\}, \qquad Vch = \mathbf{1}\{\eta=1\}\,\mathbf{1}\{h' \neq h_1\}\,\max\{0,\ p_r\,h' - \kappa_r\,y^g\}. \tag{17}$$

A homeless voucher holder does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again.

2.4 The budget constraint

$$c + p_v v + a' \ = \ y^n + a \ - \ p_r\,h'\,\mathbf{1}\{h' \neq h_1\} \ + \ Vch + F, \qquad c>0,\ v\geq 0,\ a'\geq \underline{a}. \tag{18}$$

The drug price is p_v and market rent is p_r, both as in Section 1.2.3, and a is assets carried in. Note what is absent: treatment, which costs utility, and bequests, which go to the government. How c, v and n are chosen out of it is with the rest of the optimisation, in Section 2.11.

2.5 Housing, and the homelessness loop

Housing is a four point grid. h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is the HUD point-in-time definition and excludes doubling up; it carries a small service flow and no rent, a fixed fraction of the smallest rental rather than a level of its own. The three rental sizes are pinned at both ends and spaced by a curvature between them,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h} + \left(\tfrac{1}{2}\left(\bar{h}-\underline{h}\right)^{1/\varsigma}\right)^{\varsigma}, \qquad h_4 = \bar{h}, \tag{19}$$

so three parameters describe the grid: $\underline{h}$ the smallest rental size, $\bar{h}$ the largest, and ς the curvature that places the rung between them. At $\varsigma = 1$ the ladder is uniform and at $\varsigma > 1$ the middle rung sits closer to the bottom, which is what a housing size distribution with most households in the largest category requires.

Homelessness then enters three separate equations: survival (24), the employment transitions (26) and the SUD recovery hazard (21), through $\kappa_{\phi H}(j)$ and κ_{ueH} in Section 2.8 and μ_H in Section 2.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. Mental illness is the one thing it does not reach. The SMI chain runs on its own, and that is a finding rather than an omission: the trial that housed people with serious mental illness looked for an effect on their symptoms and did not find one, while finding large effects on their housing. Section 2.7 gives it.

2.6 Drug use, the SUD, recovery and treatment

A household either abstains or uses, with v chosen continuously. A clean household that uses acquires the SUD with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda v}, \tag{20}$$

In the baseline a clean household may initiate only up to an age J_E, the last age at which use can begin.[7] A household with the SUD recovers with probability

$$\pi_r(h', m, \tau) = \min\left\{1,\ \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\right\}, \tag{21}$$

with $\bar{\pi}_r$ the baseline annual chance of recovering while housed, clean of SMI and untreated, and three multipliers on it: $\mu_H < 1$, so recovery is harder on the street, $\mu_m < 1$, harder with SMI, and $g > 1$ the treatment effect.

μ_H is how housing acts on drug use here, and it is worth being exact about what that claim is and is not. Housing does not change what a household wants: flow utility is the same function of (c, h, v) housed or not. What housing changes is the chance of getting out of the disorder once it is carried, recovery on the street being μ_H times recovery while housed. Milby et al. (2005) randomised cocaine dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies μ_H in our calibration.

Treatment τ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (14). It also costs the drug: a household in treatment must abstain in that period, $v = 0$ whenever $\tau = 1$, which is imposed as a constraint on (30) rather than priced. On the other hand, because it is not using, its mortality rate goes back to the baseline for that period.

g is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline.

2.7 Serious mental illness

m is an exogenous two state Markov chain representing SMI, whose prevalence p_m is about 5.5 percent of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is a function of age. Writing $p(j)$ for prevalence at age j and λ_m for the annual chance of leaving the state,

$$\text{onset}_m(j) = \frac{p(j{+}1) - p(j)(1 - \lambda_m)}{1 - p(j)}, \tag{22}$$

which is the accounting identity that reproduces the observed age profile exactly, backed out of the age profile of prevalence in NSDUH 2023 and the recovery hazard. The transitions are then

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m(j) \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m, \tag{23}$$

[7] Without it there is little consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it.

where $\iota > 1$ is the multiplier on onset for a household carrying the SUD, which makes a SUD raise the hazard of SMI. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter (Murrie et al., 2020), and in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74 (Nielsen et al., 2017).

SMI then affects five objects: productivity conditional on working through ρ_m in (15), both employment margins through κ_{uem} in (26), mortality through $\kappa_{\phi m}$ in (24), the SUD recovery hazard through μ_m in (21), and disability income, for which eligibility is being out of work with SMI and the replacement rate is ζ_{DI}. It does not change the utility function.

The mental health chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

2.8 What the SUD, SMI, and homelessness cost

Death risk. Survival is

$$\phi(j, h', d, m, \tau) = 1 - \min\left\{1,\ (1 - \bar{\phi}(j))\, \kappa_{\phi H}(j)^{\mathbf{1}\{h'=h_1\}}\, \kappa_{\phi d}(j)^{\mathbf{1}\{d=1\}\mathbf{1}\{\tau=0\}}\, \kappa_{\phi m}^{\mathbf{1}\{m=1\}}\right\}, \tag{24}$$

with $\bar{\phi}(j)$ the baseline survival age profile and $\kappa_{\phi H}(j)$ and $\kappa_{\phi d}(j)$ two age varying multipliers, and $\kappa_{\phi m}$ one that is not, raising the death hazard for homelessness, the SUD and SMI respectively. The SUD multiplier is switched off in a period the household is in treatment, $\tau = 1$, which is why the survival function takes τ as an argument at all. The excess mortality of the disorder is overwhelmingly a mortality of use, poisoning above all, and treatment forbids use in the period it is taken, so a household that is not using does not carry it. I make this assumption because that is what the cause specific evidence shows. The meta-analyses of mortality among people using extramedical opioids report overdose as the largest single contributor to the excess (Larney et al., 2020; Degenhardt et al., 2011), and in CDC WONDER drug poisoning alone accounts for about a fifth of all deaths at ages 25 to 44, which is the moment Section 3.3 reports the model against. $\bar{\phi}(j)$ is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$.

The homelessness multiplier cannot be read off the data directly, and the reason is worth spelling out. Baggett et al. report a *standardised mortality ratio* (SMR) for homeless adults: deaths among the homeless at a given age divided by deaths in the general population at that age, which they publish by age band and which we write $RR(j)$. It cannot be used as $\kappa_{\phi H}(j)$ because the people it is measured on are not only homeless. A share $s = 0.217$ of them also carry a SUD (Gutwinski et al., 2021), which raises mortality by itself, so the published ratio is a mixture of two effects and attributing all of it to housing would overstate what housing does.

Three things are observed and two are solved for. Observed: $RR(j)$, Baggett's ratio; $\sigma_d(j)$, the shape of the SUD's mortality effect over age; and s, the SUD share of the homeless. Solved for: $\kappa_{\phi d}(j)$ and $\kappa_{\phi H}(j)$, one age profile from each equation. Picking the single number $\bar{\kappa}_{\phi d}$ settles both: the first equation turns it into $\kappa_{\phi d}(j)$ at every age, and the second then reads $\kappa_{\phi H}(j)$ off Baggett's ratio at every age.

Undoing the mixture needs the SUD's own multiplier at each age. That is what $\sigma_d(j)$ supplies: the age profile of drug poisoning deaths from CDC WONDER 2023, normalised to one at the age where it peaks, so $\sigma_d(j)$ is the share of the SUD's peak mortality effect that is present at age j. Writing $\bar{\kappa}_{\phi d}$ for the level of the SUD multiplier,

$$\kappa_{\phi d}(j) = 1 + (\bar{\kappa}_{\phi d} - 1)\, \sigma_d(j), \qquad \kappa_{\phi H}(j) = \max\left\{1,\ \frac{RR(j)}{(1-s) + s\, \kappa_{\phi d}(j)}\right\}. \tag{25}$$

The denominator is the average mortality multiplier in Baggett's population: a share $1 - s$ of it carries homelessness alone and a share s carries the SUD as well. Dividing the observed ratio by it leaves the homelessness effect on its own.

$\bar{\kappa}_{\phi d}$ is assigned from the literature on all cause standardised mortality ratios for opioid use disorder, which report values between 6 and 10 (Degenhardt et al., 2011; Hser et al., 2017; Larney et al., 2020), and the calibration section reports the value taken. The drug price participation elasticity is not targeted; it is reported as a validation moment and compared with the range estimated in the literature, -0.3 to -1.1 (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998; Olmstead et al., 2015).

$\kappa_{\phi m}$ carries no age argument because the data give it basically none. Olfson et al. (2015) give all cause standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across ages 20 to 34, 35 to 54 and 55 to 64. Flat, if anything rising.

Unemployment risk. Job finding and separation are

$$\pi^{ue} = \min\left\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\right\}, \qquad \pi^{eu} = \min\left\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\right\}, \tag{26}$$

with $\bar{\pi}^{ue}$ and $\bar{\pi}^{eu}$ the baseline job finding and separation rates and κ_{ueH}, κ_{ued} and κ_{uem} the multipliers for homelessness, the SUD and SMI, so each multiplier below one makes work simultaneously harder to find and easier to lose.

The child's income risk. A parent with a SUD, or one who ends up on the street, lowers the productivity (y_{child}) its child is born with, through δ_d and δ_h respectively, as shown in (39) in Section 2.14. That is a risk to the household's own objective only to the extent that ω, the weight the parent places on its child's lifetime utility in (30), is positive. Since ω is small (see Section 3 for details), the household weighs a fraction ω of this risk and $1-\omega$ of it falls outside the problem entirely. It is therefore the one risk here that is principally an externality rather than a pure personal cost, which is what makes it a candidate for policy in a way that mortality and unemployment risk, both fully internalised, are not.

2.9 The voucher, and the two policies

A household without a voucher draws one with probability π_V, the same lottery probability as in Section 1.2.3, if it is income eligible, and a household with one keeps it while eligible, so unlike Section 1.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + ra \ < \ \kappa_V\, y^{\text{med}}. \tag{27}$$

with κ_V the eligibility cutoff expressed as a share of median gross income y^{med}. Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability p_{det} by a household that uses drugs or has a SUD, and with probability p_{fp} by one that did not,

$$\Pr(\eta'=1) = \Pr(\eta'=1)^{HF} \times \begin{cases} 1-p_{\text{fp}} & \text{if } (v=0 \text{ and } d=0) \text{ or } \tau=1, \\ 1-p_{\text{det}} & \text{otherwise.} \end{cases} \tag{28}$$

There are four cases and the rule covers them. A clean household that does not use meets the condition and faces only the false positive rate p_{fp}. A clean household that uses fails it and faces p_{det}. A household carrying the disorder that does not take treatment fails it and faces p_{det} as well. And a household carrying the disorder that does take treatment is abstaining, $v=0$ by the restriction in Section 2.6, so it meets the condition and faces p_{fp}. There is no fifth case: treatment is available only when $d=1$, and it forbids use, so no household is ever both in treatment and using.

This is the one place where the quantitative model changes the *logic* of Section 1.5 rather than its setting. There, conditionality is perfectly enforced and perfectly met, so Treatment First deters everyone and nobody is thrown off the rolls. Here detection is imperfect and so are households: some fail the condition because they cannot get clean, and some fail it because the test is wrong. Those two groups are the empirical case for Housing First, neither exists in the stylized model, and that is why its welfare numbers were illustration.

2.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

Productivity y. Fixed over the life cycle, so $y' = y$. I do this for two reasons. First, it is what keeps the child's penalty (39) a shift along a single scalar rather than a shift across the nodes of an age varying discretisation, which significantly reduces the computation time to manageable levels. Second, it still captures the key mechanisms, because unemployment and the penalty ρ_d are the income risks that link the SUD to housing, and those are retained in full.

Employment e. A household that loses their job receives insurance with certainty for one year. With π^{ue} and π^{eu} from (26), it gives the following transitions:

$$e = 1 \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu} \end{cases} \qquad e \in \{2, 3\} \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}, \end{cases} \tag{29}$$

State 2 is reachable only from employment and never from itself, so a household is in it for the first year out of work and in state 3, which pays nothing, from the second year on. One year is if anything generous: most states pay for twenty six weeks.

The SUD d. By (20) if $d = 0$ and (21) if $d = 1$, conditioning on the chosen v, h' and τ.

Mental illness m. By (23), conditioning on d.

The voucher η. By (27) and (28), conditioning on next period's eligibility and, under Treatment First, on the chosen v, the status d and the treatment choice τ.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (30) is taken against.

2.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot)$ given below,

$$\begin{aligned} V(j, a, y, e, d, m, \eta) = \max_{\{n,\, h',\, v,\, \tau,\, a'\}} \Bigg\{ & \frac{\big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x H \\ & + \; \beta\,\phi(j, h', d, m, \tau)\int_{\mathcal{S}} V(j{+}1)(s')\,\Gamma\big(\mathrm{d}s' \mid s, n, h', v, \tau\big) \\ & + \; \beta\,\big[1 - \phi(j, h', d, m, \tau)\big]\,\omega\,\Big[\mathbb{E}\,V\big(1, 0, y_{\text{child}}, e', 0, 0, 0\big) \\ & \qquad - \; \mathbb{E}\,V\big(1, 0, y_{\text{base}}, e', 0, 0, 0\big)\Big] \Bigg\} \end{aligned} \tag{30}$$

subject to the budget (18), the income definitions (15) to (17), the choice set of dwellings (13), $\tau = 0$ whenever $d = 0$, $n = 0$ whenever $j \geq J_R$ or $e \neq 1$, $v = 0$ whenever $\tau = 1$, $a' \geq 0$, $c \geq F$ so that consumption never falls below the food stamp floor, and $V(J{+}1, \cdot) = 0$ with $\phi(J) = 0$, so that at the last age the child is the whole continuation. The integral runs over next period's state $s' = (a', y, e', d', m', \eta')$; the second line of (30) is the household's own continuation, and the third is the child's expected lifetime utility, entered at age one with zero assets and with its own employment status e' drawn as any newborn's is.

A household that does not survive from j to $j + 1$ leaves a child, so the continuation is the survival weighted value of its own next period plus the death weighted value of what the household's own state costs

that child, each discounted, with y_{child} as defined in (39). What that buys is a second route by which the household's own decisions, to use and to go unhoused, reach its child. $\phi(j,\cdot)$ itself depends on the dwelling chosen, so the housing decision moves the weight on the two terms directly. The drug does not move it within the period, because ϕ takes the disorder as a state and v only raises the chance of carrying it *next* period. The drug reaches the child through the continuation instead, which is where the first order condition below picks it up.

The third line is a *difference* and not a level, and the distinction is not cosmetic. Written as a level, ω times the child's utility, the term is worth more to a household the likelier it is to die, because it is the death branch that carries it. A SUD raises the death hazard several fold, so a level would pay most to exactly the households the paper is about, and it would do so for reasons that have nothing to do with the child.

Subtracting the value of a child born to a clean, housed parent removes that. The term is zero for a household in the reference state and negative otherwise, so what enters the decision is only the loss the parent's own state imposes on its child, weighted by the chance that child is born at all.

At $\omega = 0$ the damage a household does to its child is entirely external; at $\omega > 0$ the household internalises a share ω of it, and only $1-\omega$ remains external. Because the term is a difference, ω is the weight on that damage and nothing else. Since the child's productivity depends on whether the parent carried a SUD or was homeless, and both depend on v and h', the child term in (30) is what puts the child inside the parent's drug and housing decision.

How the choices separate, and where they do not. Taking hours, the dwelling and next period assets as given, what is left to divide between consumption and the drug is

$$x \;=\; y^n + a \;-\; p_r\, h'\, \mathbf{1}\{h' \neq h_1\} \;+\; Vch + F \;-\; a', \tag{31}$$

so the budget (18) reads $c + p_v v = x$ and the whole of the remaining problem is one number: how to split x between c and v, since either one determines the other.

Were the drug an ordinary good, one whose only consequence is the flow utility it delivers this period, that split would follow from flow utility alone. It does not, because v enters the transition as well as the utility: by (20) another unit of the drug also buys a higher chance of waking up next period with the disorder. So there are three terms in the first order condition for v, not two,

$$\underbrace{\frac{\theta}{b+v}\,\Lambda}_{\text{value of another unit}} \;=\; \underbrace{\frac{p_v\,(1-\gamma-\theta)}{c}\,\Lambda}_{\text{the consumption it displaces}} \;+\; \underbrace{\lambda e^{-\lambda v}\Big[\,\mathbb{E}V(\cdot \mid d'=0) - \mathbb{E}V(\cdot \mid d'=1)\,\Big]}_{\text{the risk it adds}}, \tag{32}$$

where $\Lambda = \big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}$ is the Cobb-Douglas bundle of (14). It multiplies the first two terms and not the third, because the third is a difference of continuation values and is therefore already in utils. Were it absent, Λ would divide out of the condition entirely and the split would reduce to the usual Cobb-Douglas one; it is the third term that stops it cancelling, and that is why the marginal utility level matters here and not in an ordinary two good problem. That third term is the marginal hazard $\pi_d'(v) = \lambda e^{-\lambda v}$ times what the disorder costs, the gap between the two continuation values. That gap is where the externality on the child enters the drug decision, and it is why there is no fourth term. Each continuation value carries its own child term, at $d'=0$ and at $d'=1$, so the difference between them already contains the ω share of what the disorder costs the child alongside what it costs the household itself. A fourth term would be needed only if using raised the chance the child's term is realised within the period, and it does not: ϕ depends on the disorder as a state, not on how much is bought today. This is Section 1's mechanism inside the quantitative model: because π_d rises in v, how much to use is not decided on the budget alone.

There are two cases in which the third term is zero, and in both the household faces no such trade off: one that abstains, where $v = 0$, and one that already carries the disorder, whose recovery hazard (21) does not depend on how much it uses. For both the condition is the Cobb-Douglas split in closed form,

$$c = \max\Big\{F,\; \frac{1-\gamma-\theta}{1-\gamma}\big(x + p_v b\big)\Big\}, \qquad v = \max\Big\{0,\; \frac{x-c}{p_v}\Big\}, \tag{33}$$

whose corner is the abstention decision: a household buys nothing when the Stone-Geary endowment already exceeds what it wants, which is how a large b deters use and is the threshold of Section 1.2.4. The outer

maximum on c is the food stamp floor. Where it binds the Cobb-Douglas split no longer describes the allocation: consumption is held at F and the drug takes what the budget leaves, so the floor changes behaviour and not only welfare. Where the third term is present the answer lies between (33) and abstention, since risk can only push towards less, so it is found by a one dimensional search on that interval.

Hours then solve

$$u_c(c)\,\frac{\partial x}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}, \tag{34}$$

and the clawbacks in (17), at a rate of one for food stamps and κ_r for the voucher, make $\partial x/\partial n$ jump *up* at the hours where each transfer runs out, because the clawback stops. That is the welfare trap, and it is the channel through which conditionality changes labour supply rather than only consumption.

2.12 The government

The government pays social security, unemployment insurance, disability, food stamps and vouchers, and supplies a public good $\bar{G}$ that does not enter private consumption and is held fixed across simulations, so that policies are compared at a constant level of it. It finances this with the two taxes in (16) and the estates of the deceased, with the level τ_ℓ clearing the budget in every simulation,

$$\underbrace{\textstyle\sum y^{g} - \tau_\ell \sum (y^{g})^{\xi} + \tau_k \sum ra + \text{estates}}_{\text{revenue}} \;=\; \underbrace{SS + UI + DI + F + Vch + \bar{G}}_{\text{outlay}}. \tag{35}$$

In the stylized model the two policies happen to cost the same and the comparison is unaffected. Here a policy that houses more households spends more, τ_ℓ rises to pay for it, and the higher tax lowers the after tax return to an extra hour: it enters (34) through $\partial x/\partial n$, which is net of the tax schedule in (16), so hours fall and saving with them, for everyone, including the households the programme never reaches, and it runs against Housing First.

2.13 The drug market

Drug production has cost $C(S_v) = \frac{\nu_1}{\nu_2} S_v^{\nu_2}$, where S_v is the aggregate quantity. Supply satisfies

$$p_v = \nu_1 S_v^{\nu_2 - 1}, \qquad S_v = \int v(s)\,\mathrm{d}\Phi(s), \tag{36}$$

with price elasticity of supply $1/(\nu_2 - 1)$ and ν_1 normalised so that the quantity supplied equals the quantity demanded at the benchmark price, which is normalised to one. The supply elasticity, its range and the evidence behind them are set in Section 3.

2.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi(j,\cdot) \int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\,\mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{37}$$

together with the newborn inflow

$$\Phi_1(\mathcal{S}') \;=\; \int \big[1 - \phi(j,\cdot)\big]\,\mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_j(s) \;+\; \int \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_J(s), \tag{38}$$

where $\mathcal{N}$ places a newborn according to (39), so that the mass of newborns equals the mass of the dead exactly and the two equations together conserve the population.

Here $\phi(j,\cdot)$ is evaluated at the household's own $(h'(j,s), d, m)$. The mass that does not survive, plus the mass at $j = J$, is replaced by newborns at $j = 1$ with zero assets, clean, without SMI and without a

voucher, and subject to the same unemployment shocks as anyone else, whose productivity is drawn from the unconditional distribution Ω_y of permanent productivity across households scaled by the transmission penalties δ_d for a child whose parent carried the SUD and δ_h for one whose parent was homeless,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}. \tag{39}$$

These two penalties carry the paper's central mechanism, so it is worth setting out the evidence for each of the three links they compress rather than asserting the reduced form.

A parent's disorder changes the childhood. It does so at scale and through a channel that is measured: parental alcohol or drug abuse is recorded as a circumstance at removal for 38 percent of children entering out of home care in 2025, against 18.5 percent in 2000, and in FY2017 that was 96,400 children, or 36 percent of all removals (AFCARS). Whether removal itself helps or harms is contested, and the paper does not need to take a side: Doyle (2007) finds that children on the margin of placement do better left at home, while Gross and Baron (2022) find that placement improved safety and schooling. δ_d is a reduced form for the whole bundle of that childhood, not for removal alone.

A parental shock transmits to the child's own earnings. This is the step the reduced form actually assumes, and it is estimated outside the substance use literature entirely: children whose fathers were displaced by a firm closure earn about 9 percent less as adults (Oreopoulos, Page and Stevens, 2008). For the substance channel specifically, Nilsson (2017) uses a Swedish experiment that sharply raised alcohol availability and finds substantially worse adult earnings and schooling among the cohorts exposed in utero, though that is prenatal exposure rather than the childhood environment δ_d describes.

Housing is one of the environments that does the transmitting. Children displaced by Chicago's public housing demolitions were 9 percent more likely to be employed as adults and earned 16 percent more a year, and those who moved before thirteen earned $3,036 more at age 26 (Chyn, 2018); Chetty et al. (2016) put the gain from a move to a lower poverty neighbourhood before thirteen at 31 percent, which is the figure δ_h is set to. On the adult side, eviction lowers earnings and raises homelessness in linked administrative data for Cook County and New York City (Collinson et al., 2024).

2.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{n_t(j,\cdot),\, h'_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{\tau_{\ell t},\, p_{vt},\, \bar{y}_t,\, y_t^{\text{med}},\, H_t\big\}_{t=0}^{\infty}$,

such that, at every date t,

1. the policy functions solve (30) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the newborn rule (39) and (38), and the mass of newborns equals the mass of the deceased;
3. the government budget (35) holds at $\tau_{\ell t}$;
4. the drug market clears at p_{vt}, so (36) holds with aggregate drug demand $S_{vt} = \int v_t(s)\,\mathrm{d}\Phi_t$;
5. the public good is supplied at its assigned level, $G_t = \bar{G}$;
6. $\bar{y}_t$, median gross income y_t^{med} and the homelessness rate H_t are the aggregates implied by Φ_t.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the policy comparisons report. The sequence version is stated for completeness, and because the transition exercise in the results needs it.

3 Calibrating the quantitative model

The model has fifty five parameters, some of which vary in value by age. Thirty three are calibrated externally, from data or from the literature, and twenty two are calibrated internally, inside the model. Four further parameters govern drug testing and detection; they are inert in the benchmark, where nothing is conditional, so they are set with the Treatment First experiment rather than here. They are split across the two tables of this section: thirty two are the main story of this paper and its policy analysis (Table 5, sixteen internally and sixteen externally calibrated), and twenty three are standard (Table 6, five internally and eighteen externally).

The tables carry the parameter, its description, the moment and source used to fix its value, and the value itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and why its moment identifies it.

3.1 The thirty two parameters that carry the story of this paper

Twenty two of the fifty four are the furniture of any life cycle model, whereas the rest are specific to this paper, and all thirty two of those carry its main story and policy analysis. They are set out in Table 5, with the parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table.

The comparison between Housing First and Treatment First has a shape, and every entry below is read against it, so it is worth setting out this comparison once and in full before the entries begin.

Three things decide the ranking between Housing First and Treatment First. The **private value** is what a household itself gains from being housed and from not carrying a disorder: consumption, a dwelling rather than the street or shelter, and above all the years of life that homelessness and the disorder take away. The **externality** is what lands on somebody else: on the child, who inherits not the disorder or homelessness but a worse productivity draw associated with the parent's SUD and with the parent's homelessness, which δ_d and δ_h price separately. This means that the child is likelier in turn to be poor, to use, to acquire a disorder and mental illness as a consequence, to be unhoused and to die young. The externality also lands on everybody, through the disamenity of aggregate homelessness. The parent internalises only ω of the part that falls on its child, and ω is small, which is what leaves something for a policy to correct. And the **distortion** is what the programmes cost in behaviour, and it comes from two places. The programmes create distortions due to their design: assistance is means tested and the rent share is clawed back at κ_r, so a recipient who works more loses eligibility or part of the transfer, which requires even higher taxes on everyone else to fund even larger programmes. And assistance is paid for by distortionary taxes, which lower the after tax return to an hour for households neither policy ever reaches.

Housing First is redistribution, removes externalities, but creates another. Mechanically it removes two externalities: a child of a housed parent does not carry the penalty δ_h prices, and aggregate homelessness falls, which every household values through κ_x. However, it creates one at the same time, because assistance that arrives whatever a household does insures it against the consequences of using, so the SUD rate rises and reaches the child through δ_d.

Treatment First does not redistribute, removes an externality but creates two others. Curing the parent is how the child's worse draw is avoided. But conditioning assistance means some households do not receive it and end up homeless, especially in the short run, and over that horizon it creates the two externalities Housing First removes. The short run matters on its own account: a household that wants to stop cannot do it immediately, and the transition is costly while it lasts, which is why the policy analysis reports the transition and not only the steady states.

And general equilibrium returns part of it. First, the drug price is endogenous, set by the supply curve that ν_1 and ν_2 describe, so curing disorders lowers drug demand, lowers the price, and makes using cheaper for those still deciding: part of what treatment removes comes back through lower prices. Second, when the government spends more on either policy and the tax rises, hours fall, the base shrinks and the tax has to rise again. Third, Treatment First has an offset of its own in the other direction: sending more households

into treatment raises the recovery rate faced by everyone deciding whether to use, and those consequences affect a newborn when it makes its own decisions, so the policy changes behaviour beyond the households it treats. And because κ_x is levied on the aggregate headcount, any movement in homelessness feeds back into everyone's utility. So the mechanical effects above are upper bounds on both sides.

For every parameter the things worth knowing are what happens to the private value, to the externality and, where it applies, to the distortion, *as its value rises or falls*, and each entry says so in a line before giving the moment. The middle blocks are where the two move together: because the child inherits a higher chance of landing in exactly the states those parameters price, anything that raises the private cost of using or of being unhoused raises the externality as well. Both policies work on those blocks, from opposite ends.

Table 5: The thirty two parameters that carry the story of this paper

Parameter	Description	Set	Moment and source	Data	Model	Value
Who uses drugs						
θ	Weight on drugs in utility	Int.	Drug spending over income, users, NSDUH 2024	35%	—	—
b	Whether a household uses	Int.	Drug participation, NSDUH 2024	5.13%	—	—
λ	SUD hazard per unit used	Int.	Adult SUD rate, NSDUH 2024	3.68%	—	—
$\bar{\kappa}_{\phi d}$	Increased mortality with a SUD	Ext.	Mortality ratios, Hjorthøj et al. (2023)	6 to 10	n.a.	8.00
$\aleph$	Flow value of being alive	Int.	Value of a statistical life, Viscusi (2003)	\$7.5m	—	—
What using and being homeless cost						
ρ_d	Earnings lost to a SUD	Int.	Employment rate of the homeless, HUD 2023	50%	—	—
κ_{ued}	Employment multiplier, SUD	Int.	Unemployment rate given a SUD, NSDUH 2024	16.25%	—	—
h_1	Service flow of homelessness	Int.	Homelessness rate, HUD 2023	0.399%	—	—
$\kappa_{\phi H}(j)$	Increased mortality when homeless	Ext.	Mortality ratios by age, Baggett (2013)	9.1 to 1.1	n.a.	4.41 to 1.09
κ_{ueH}	Employment multiplier, homeless	Ext.	Desmond and Gershenson (2016)	0.80	n.a.	0.80
ζ_{DI}	Disability replacement, SMI	Ext.	SSDI and SSI rates, SSA 2023	0.25	n.a.	0.25
$\underline{c}$	Food stamp consumption floor	Int.	Food stamps over labour income, USDA	0.42%	—	—
$\bar{\pi}_V$	Voucher lottery probability	Int.	Recipients over eligible, HUD 2023	26.6%	—	—
The mental illness chain						
ι	SUD multiplier on SMI onset	Ext.	Danish registers, Nielsen et al. (2017)	2.00	n.a.	2.00
λ_m	Chance of recovering from SMI	Int.	SMI share of homeless, Gutwinski et al. (2021)	25%	—	—
ρ_m	Earnings lost to SMI	Int.	Labour income of the homeless relative to average	19.1%	—	—
κ_{uem}	Employment multiplier, SMI	Int.	Employment rate given SMI, NSDUH 2023	30%	—	—
μ_m	SUD recovery multiplier, SMI	Ext.	Dual diagnosis relapse, Bradizza et al. (2006)	0.82	n.a.	0.82
$\kappa_{\phi m}$	Mortality multiplier, SMI	Ext.	Olfson et al. (2015)	2.50	n.a.	2.50
Recovery and treatment						
$\bar{\pi}_r$	Baseline SUD recovery rate	Ext.	Ten year remission rate, Vaillant (1995)	0.025	n.a.	0.025
μ_H	Recovery multiplier when homeless	Int.	Weeks abstinent, Milby et al. (2005)	1.473	—	—
g	Treatment effect on recovery	Ext.	Odds ratio, Milby et al. (2005)	1.35	n.a.	1.35
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take up, NSDUH 2023	24%	—	—
Externalities and distortions						
ω	Weight on the child	Int.	Use rise with altruism off, NSDUH 2024	18.3%	—	—
δ_d	Child's earnings, parent SUD	Ext.	Child SUD ratio, McGovern et al. (2023)	2 to 4	n.a.	0.30
δ_h	Child's earnings, homeless	Ext.	Poverty exposure on earnings, Chetty et al. (2016)	31%	n.a.	0.31
κ_x	Disamenity of homelessness	Ext.	Willingness to pay, Loubière et al. (2020)	0.20%	n.a.	0.0008
ω_ℓ	Weight on leisure	Int.	Average hours worked, Guvenen et al. (2023)	0.40	—	—
σ_ℓ	Curvature on leisure	Ext.	Frisch elasticity of 0.5, Chetty (2012)	0.50	n.a.	3.00
The drug market						
ν_1	Drug supply shifter	Ext.	Normalisation, so $p_v = 1$	norm.	n.a.	norm.
p_v	Price of drugs	Ext.	Clears the drug market in the benchmark	1.00	n.a.	1.00
ν_2	Drug supply curvature	Ext.	Elasticity 2; Hansen (2017)	2.00	n.a.	1.50

Note: n.a. in the Model column marks a parameter that is set from outside rather than calibrated inside the model. For those the model produces no internal target to compare with the data, so the cell reads n.a. rather than a dash, which keeps them distinct from the internally calibrated parameters.

Who uses drugs. These four set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

θ, the weight on the drug in utility. How much a household wants the drug at all, against consumption and housing. It is the level of the temptation where b is its curvature, and it is the parameter the amount spent on drugs, when it does, turns on. As it rises use rises on the extensive and intensive margins, so the private cost and the externality both rise and there is more for either policy to act on.

It is calibrated internally to drug spending over income across users, 35 percent in NSDUH 2024. The moment is built in two steps, because users are not one group. Households that buy without carrying a disorder and households that carry one have very different ratios, so each group's own total drug spending is divided by that group's own total income, and the two ratios are then averaged with their shares of the using population as weights:

$$\text{drug share} \;=\; \frac{\mu_e}{\mu_e+\mu_d}\,\frac{p_v\bar{v}_e}{\bar{y}_e} \;+\; \frac{\mu_d}{\mu_e+\mu_d}\,\frac{p_v\bar{v}_d}{\bar{y}_d},$$

where $\bar{v}_g$ and $\bar{y}_g$ are the average drug quantity and average income in group g, and μ_e and μ_d are the masses without and with the disorder. The target follows from the two group ratios and the split between them: users without a disorder spend 11 percent of income on the drug and users with one spend 44, and the share of users carrying a disorder is the SUD rate over the participation rate, $0.0368/0.0513 = 0.717$, so the average is $0.717 \times 0.44 + 0.283 \times 0.11 = 0.35$.

b**, whether a household uses at all.** The marginal value of the drug goes as $\theta/(b+v)$, so a larger b makes the first units worth less and a household with little to spend buys nothing. As b rises use falls, so the private cost and the externality both fall, and housing assistance is less likely to be translated into more spending on drugs: b is the size of the moral hazard. Its value is chosen so the model generates a drug participation rate of 5.13 percent, the NSDUH 2024 figure for adults aged 26 and over, which identifies it sharply because participation is close to a threshold function of b.

λ**, the hazard per unit used.** This is what makes using *more* risky, and it is the step that turns spending into a disorder. As λ rises both the private cost and the externality rise, and Housing First's moral hazard costs more, since it is λ that converts an extra voucher into a higher SUD rate Treatment First attempts to cure. Its value is chosen so the model generates the share of adults carrying a substance use disorder, 3.68 percent among those aged 26 and over in NSDUH 2024. That is the level rather than the conversion rate, and it is the level because the two published numbers do not agree with each other: a conversion of 33.2 percent applied to a participation rate of 5.13 percent implies a disorder rate near 1.7 percent, not 3.68. The disorder rate is the better measured of the two, so it is the one the hazard is matched to and the conversion the model implies, 71.7 percent, is reported rather than imposed.

$\bar{\kappa}_{\phi d}$**, increased mortality with a SUD.** This is the parameter the whole paper turns on. Mortality truncates the value of life for a user, which hits households with the highest life values hardest and thus reverses the income gradient, making the poor the users rather than the rich. As it rises the private cost rises, and the externality rises with it, since a child born to a parent with a SUD has a lower value of life. Both policies reduce these deaths, treatment by curing and housing because recovery is less likely when homeless. It is calibrated externally, at the midpoint of the standardised mortality ratios for SUD in Hjorthøj et al. (2023), which run six to ten, with the age profile from CDC WONDER 2023.

$\aleph$**, the flow value of being alive.** The other half of the mortality mechanism: $\bar{\kappa}_{\phi d}$ sets how much a SUD raises the chance of dying and $\aleph$ sets what the life being lost is worth. As it rises the private value of every life either policy saves rises, and use falls, so the externality falls. Its value is chosen so the model generates a value of a statistical life of \$7.5m, from Viscusi and Aldy (2003).[8]

What using and being homeless cost, and the voucher. Both policies work on this block, from opposite ends: each parameter says how bad a state is, so each sets what a voucher is worth to a household in it and what being cured of the reason for it is worth. The voucher itself closes the block, since it is the instrument that acts on all of them. Because the child inherits a higher chance of landing in these states, everything here moves the private value and the externality together.

ρ_d**, earnings lost to a SUD while still working.** It reduces the value of life when a household acquires a SUD, since the earnings it costs would have been enjoyed for the rest of that life. It also does a job nothing else can: it lets an *employed* household be poor enough to be priced out of housing, which an employment effect alone cannot deliver, since that makes essentially every homeless household jobless,

[8] I check that no household has a negative value of life in equilibrium, so death is never an optimal outcome.

whereas about half are not in the data. As ρ_d rises the penalty shrinks, so the private cost falls and, because the impact on the child is less harmful, the externality declines. These earnings are lost whether the household is housed or homeless, so treatment affects them and housing does not.[9] Its value is chosen so the model generates the employment rate of the homeless, 50 percent, from HUD's 2023 point-in-time count and TEDS-A 2022.

κ_{ued}**, the employment multiplier for a SUD.** The same channel on the extensive margin: regaining employment is less likely and losing it more likely with a disorder. As it rises the private cost rises, even more than for ρ_d, the externality rises with the child's draw, and the tax base narrows since the household does not work and receives unemployment insurance, so it costs taxpayers as well. Its value is chosen so the model generates the unemployment rate given a SUD, 16.25 percent in NSDUH 2024.

h_1**, the service flow of homelessness.** It sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is one for which the mortality cost of using is small. As h_1 rises the street is less bad, so the private cost of homelessness falls, a voucher is worth less, and the externality on a child likelier to be unhoused falls with it. Its value is chosen so the model generates the homelessness rate, 0.399 percent of households in HUD's 2023 point-in-time count.

$\kappa_{\phi H}(j)$**, mortality when homeless.** As it rises the private cost of being unhoused rises and thus so does the private value of a voucher. Baggett et al. (2013) give race adjusted mortality ratios for homeless adults of about 9.1 at ages 25 to 44, falling to 1.1 at 65 to 84. Those are totals for a population most of which also carries a SUD, so it is set externally age by age net of that composition, which is why it is derived jointly with $\bar{\kappa}_{\phi d}$ rather than read off directly, as set out in Section 2.8 at (25).

κ_{ueH}**, employment lost to homelessness.** Here the homelessness itself costs the job, so housing mechanically recovers the earnings and treating does not. As it rises the private cost and the externality rise together, which makes it the clearest case in the table of a parameter loading Housing First's side, and it also reduces the tax burden since these households are more likely to be employed. It is set externally at 0.80, from Desmond and Gershenson (2016).

ζ_{DI}**, the disability transfer for the non employed with SMI.** It is the floor for one of the worst states in the model, being ill and out of work, so it sets how far a household can fall. As it rises the private cost of that state falls, and the externality on a child likelier to reach it falls, at the price of a higher tax on everybody else. It is set externally at 0.25, from SSDI and SSI replacement rates, SSA 2023.

$\underline{c}$**, the food stamp consumption floor.** The floor under consumption, and it belongs here rather than among the standard parameters. As it rises the private cost of every bad state falls, so a voucher is worth less at the margin and the externality on a child likelier to reach those states falls too, while the distortionary taxes that fund it rise. It is calibrated internally to food stamp spending over labour income, 0.42 percent, from the USDA.

π_V**, the chance of drawing a voucher.** This is the policy dial itself. As it rises the private gain rises and two externalities fall mechanically, the child's penalty for growing up unhoused and the aggregate disamenity, while the third rises through the SUD rate and the tax rises with the bill: the whole trade in one parameter. Its value is chosen so the model generates the 26.6 percent ratio of voucher recipients to the income eligible population in HUD's Picture of Subsidized Households, 2023.

The mental illness chain. This block is why Treatment First reaches further than the disorder it treats: a SUD raises the chance of mental illness, mental illness lowers earnings and employment and raises death, and the child is susceptible to all of this. Everything here therefore moves the private cost and the externality in the same direction, and the first three decide whether the chain exists at all.

ι**, the SUD multiplier on SMI onset.** The hinge of the whole block. Without it mental illness is exogenous to everything a policy does and treating SUD cannot touch it. As it rises a SUD causes more mental illness, so the private cost of using rises, and the externality rises further, because the child of

[9] One subtlety is worth stating, because it may seem like a contradiction with Section 1.1. There the proportional income penalty was shown to matter little for *who* uses. That is a statement about the calibrated model, not the parameter: raising ρ_d with b held fixed does change who uses, but b is recalibrated to the participation rate each time the penalty changes, and once it is, the composition of users is back where it started.

a user now inherits a higher chance of illness on top of the productivity penalty. Treatment First's return rises with them, because curing the disorder now prevents the illness too. It is set externally at two, from the Danish registers in Nielsen et al. (2017).

λ_m**, the annual chance of recovering from SMI.** The persistence of the state, and the one piece of the chain the data cannot hand over, since prevalence pins the stock and not the flows. As it rises mental illness is less persistent, so the private cost falls, because fewer years are spent ill, and the externality falls with it, since a child who is likelier to acquire a SUD and through it SMI has a higher value of life. It is calibrated internally to the SMI share of the homeless in Gutwinski et al. (2021), 25 percent.

ρ_m**, earnings lost to SMI.** The counterpart of ρ_d for mental illness, and it reaches the comparison through ι: a household carrying the disorder faces onset at ι times the baseline rate, so treatment reduces mental illness in a roundabout way and recovers these earnings for households that never fall ill. As it rises the penalty shrinks, so the private cost of illness falls and, because the child's draw improves with it, the externality falls too. Its value is chosen so the model generates the average labour income of the homeless relative to the average, 19.1 percent.

κ_{uem}**, the employment multiplier for SMI.** The same as ρ_m but on the extensive margin, and where the chain does most of its work, since employment is what mental illness costs most. As it rises the private cost rises, because being ill now costs the job as well as the wage, and the externality rises with the child's draw; the tax base narrows, which is a distortion on top of both. A child of a treated parent is less likely to carry SMI, which raises its earnings and lowers its chance of being unhoused, so part of Treatment First's return never appears in the treated household's own drug use. Its value is chosen so the model generates the employment rate given SMI, about 30 percent, the midpoint of the range for high disability diagnoses in NSDUH 2023.

μ_m**, the SUD recovery multiplier for SMI.** The second direction of the chain: not only does a SUD cause mental illness, mental illness means the SUD is less likely to be cured. As it falls the disorder is harder to leave, so the private cost of acquiring one rises and the externality rises with it, and treatment has more to overcome in exactly the households it most wants to reach. It is set externally at 0.82 from dual diagnosis relapse rates in Bradizza et al. (2006), and it is one of the three values in the paper carrying no real discipline.

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, and what makes it matter for welfare, not through income. As it rises deaths among the mentally ill rise, so the private cost rises, and the externality rises with it, because a child born to a parent with a SUD is more likely to acquire a SUD and through it SMI, and therefore more likely to die. This is how a SUD affects mortality indirectly, through SMI, rather than only by the direct route. It is set externally at 2.5, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Recovery and treatment. Both policies work on this block, since recovery is what Treatment First buys directly and what Housing First buys by housing people. The first two set how much recovery there is to buy and the last two how much of it a policy can reach.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** The hazard of getting out of the disorder for a housed household, which is the level everything else in this block scales. As it rises disorders end sooner on their own, so the private cost of acquiring one falls and the externality falls with it, and there is less for either policy to buy. It is set externally from the ten year remission rate in Vaillant (1995), which gives an annual hazard of 0.025.

μ_H**, SUD recovery when homeless.** How much less likely recovery from the disorder is for an unhoused household than a housed one. It matters more than the level $\bar{\pi}_r$ because it is a *gradient*: it is what a voucher buys beyond shelter. As it falls the gradient steepens, so housing buys more recovery: the private value of a voucher rises and the externality falls. Near one, assistance is a transfer and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery housing produced in Milby et al. (2005), a randomised trial on cocaine dependent homeless adults: 6.60 weeks abstinent in the housed arm against 4.48 homeless, so housing raises recovery by 47.3 percent.

g**, the treatment effect on recovery.** Treatment First's instrument, and the parameter its case rests on most directly. As it rises treatment works better, so a household that carries the disorder is more easily

cured, and therefore the private cost falls, since the household spends less of its life carrying one. The externality falls with it, and by the same route: a child who acquires a disorder is also more easily cured, so what its parent hands on costs it less of its own life. It is set externally at 1.35, the odds ratio in Milby et al. (2005).

$\bar{\tau}^u$**, the utility cost of treatment.** Take up is a choice, not a rule, and this is what makes households decline treatment they might gain from in the future. As it rises fewer take treatment, so the externality rises, since more people remain with a SUD and thus more children are born with lower incomes, which leads to higher SUD incidence, and the private cost rises too, since treatment is costly. It is calibrated internally to treatment take up among adults with a SUD, 24 percent in NSDUH 2023.

Externalities and distortions. The first three are what a parent's disorder and a parent's homelessness cost the child, and the fourth is what the homelessness of strangers costs everybody. All four are externalities in the strict sense: one household's choice enters another household's utility directly, and it would still do so if the programmes were paid for with a lump sum tax. The last two are not externalities but distortions, and they close the block for that reason: they say what it costs to raise the revenue either programme needs rather than whose utility a choice lands in. This is where the ranking is decided, since the private comparison already favours Housing First.

ω**, the parent's weight on its child.** A parent bears ω of what its own state costs its child, and ω is small. As it rises the parent already counts more of the child's loss, so the externality left for a policy to correct is smaller and both policies are worth less. Identifying it needs care, because *every* household here has a child, so there is no parent against non parent comparison to run inside the model. The moment is a counterfactual instead. In NSDUH 2024 past month use among parents is 0.845 of use among non parents, with parents reweighted to the non parents' joint distribution of age, income, sex and marital status so the gap is behaviour rather than composition. The child term of (30) is the only thing that makes a household in the model use less than it otherwise would, so its value is chosen so that removing the child's penalty at the calibrated parameters raises drug use by 18.3 percent, which is that same ratio read the way the model reads it: the counterfactual raises use from the parent level, so the rise is $1/0.845 - 1$ rather than $1 - 0.845$. It is the penalty that is removed and not the child, for the reason Section 2.11 gives: a term carrying the child's whole value would reward dying, and removing it would lower use rather than raise it.

δ_d**, what the child loses to a parent's SUD.** This is Treatment First's case, very nearly in full. The child does not inherit the disorder, only a worse productivity draw, and the draw is what makes it likelier to be poor, to use, to acquire a disorder and mental illness in consequence, and to be unhoused. As it rises the externality rises and Treatment First's case with it, while the parent's own private cost is unchanged, which is exactly what makes it an externality. No paper estimates the effect of a parent's SUD on a child's lifetime earnings directly, which is why the penalty is calibrated to the disorder ratio rather than lifted from an estimate. What the literature supplies is the two halves of the chain: Felitti et al. (1998) link household dysfunction to adult outcomes, and Chetty et al. (2016) measure what childhood environment does to adult earnings. It is set at 0.30, the value that puts the stylized model's ratio of child disorder rates between parents with and without a disorder in the middle of the range McGovern et al. (2023) report, who find children of a parent with a SUD two to four times likelier to acquire one. It is set rather than calibrated inside the quantitative model because that ratio needs the parent's type carried through the child's whole life, which would add a dimension to the state space for one moment; the stylized model of Section 1 is two periods long and carries it for free. Results are reported across the range of δ_d rather than at the point value alone.

δ_h**, what the child loses to homelessness.** The externality Housing First removes, working through the same compounding draw as δ_d. As it rises the externality Housing First removes grows, again leaving the private cost unchanged. Chetty et al. (2016) find that moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent; δ_h is calibrated externally to that figure, taking the earnings gain from leaving poverty as the mirror of the loss from growing up without stable housing.

κ_x**, the disamenity of homelessness.** The only term through which one household's housing status enters

the utility of another who is not its own child, which is what makes the homelessness rate a genuine fixed point rather than accounting. As it rises the externality from the aggregate headcount grows, which loads whichever policy houses more people. Setting it high therefore favours Housing First, so the higher figure is taken as the baseline and a ranking that still favours Treatment First survives the assumption least kind to it. It is calibrated externally from the contingent valuation in Loubière et al. (2020): €57 a year among the 42 percent of respondents willing to pay anything, which is 0.20 percent of average annual earnings, against €23 and 0.08 percent averaged over all respondents.

ω_ℓ**, the weight on leisure.** This sets how much labour supply responds to the tax and to the programme's own design. As it rises hours are more responsive, so the tax that funds the vouchers or the monitoring costs more, and the distortion grows against whichever policy costs more. The channel also runs back: a household that uses is likelier to be out of work and to need assistance, so its choice raises the tax everybody else pays. That feedback is a cost that travels through the government budget rather than through anybody's utility directly, which is why it is counted here: raise the same revenue lump sum and it disappears, while the child's draw and the headcount stay exactly where they were. Its value is chosen so the model generates average hours worked, from Guvenen et al. (2023).

σ_ℓ**, the curvature on leisure.** The curvature that turns ω_ℓ into an elasticity: together they decide how far hours actually move when the after tax wage does. It belongs here because the size of the distortion is exactly what this parameter sets: at a high curvature hours barely respond and the tax that funds either policy is close to a lump sum, at a low one the same revenue costs far more in forgone work. As it rises labour supply is less elastic, so the distortion falls and both policies look cheaper, without touching the private value or the externality at all. It is set externally to a Frisch elasticity of 0.5, from Chetty (2012).

The drug market. Three parameters describe the supply of the drug. Two are normalisations and the third is a supply elasticity, and they matter for one thing: they make the drug price endogenous, so a policy that changes the number of users changes what the drug costs. That is the general equilibrium offset named above, and it exists only because these are here. It is a third kind of thing again, neither an externality nor a distortion but a price: it changes the size of both without being either, and it is the one channel through which a policy reaches households it never touches without anybody's utility or the government's budget being involved.

ν_1**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark. It normalises units, not behaviour. It is a normalisation, so nothing turns on its level: it is chosen so the benchmark price is one, and moving it moves neither object.

p_v**, the drug price.** One at the benchmark by that normalisation, and free in every counterfactual. That is what lets a policy which changes how much is used move the price, and so feed back into every other household's decision. As it rises using is dearer, so use falls and both objects fall with it. It is an equilibrium object rather than a dial, which is what lets a policy that changes the number of users change what the drug costs.

ν_2**, the supply elasticity.** Sets the price elasticity of supply at 2, the middle of a range from about 0.7 to about 4. There is no direct estimate for an illicit drug, so it is bracketed by the closest legal markets: the Washington State marijuana reform implies roughly 0.79, the cigarette and gasoline literatures nearer 3.6. As it rises supply is more elastic, so the price moves less when demand does, and the general equilibrium offset to treatment shrinks: at a very elastic supply, curing households barely lowers the price and almost none of the gain comes back.

3.2 Standard parameters

The rest are conventional, recorded so the calibration can be reproduced rather than because anything turns on them. Table 6 carries each with its source and vintage, in this order. There are twenty three of them. Five are calibrated inside the model, against a moment, because there is no source to read them off; the other seventeen are set from outside, from an estimate or from statute. Each is taken below in the order of the table, with the moment or the source that fixes it, and with the value of that moment in the data.

Table 6 carries the same numbers in its Data column, against what the model produces in the Model column, which is reported only for the five that are calibrated internally since the other seventeen have nothing to reproduce.

They are not inert in the experiments, and it is worth saying so plainly. The discount factor sets how heavily a death thirty years away is weighed, the dispersion of permanent income decides how many households are poor enough to be at risk, and risk aversion decides what insurance is worth: each of them moves the private value and, through the child's draw, the externality as well. What separates them from the thirty two above is not that they do nothing but that they are not specific to this paper. They are calibrated the way any life cycle model calibrates them, to the same moments and from the same sources, so a reader who disagrees with one of them is disagreeing with a literature rather than with anything argued here. The parameters in Section 3 are the ones where this paper makes a choice.

Table 6: Standard parameters

Parameter	Description	Set	Moment and source	Data	Model	Value
γ	CD weight on housing	Int.	Renters above 30 pct of income, ACS 2023	49.7%	—	—
β	Discount factor	Int.	Wealth over output, BEA fixed assets	3.00	—	—
$\underline{h}$	Smallest rental size	Int.	Share in smallest unit size, ACS 2023	13.2%	—	—
$\bar{h}$	Largest rental size	Int.	Share in largest unit size, ACS 2023	62.7%	—	—
ς	Curvature of the ladder	Ext.	Set; the two ends carry the distribution	1.80	n.a.	1.80
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	4.85%	—	—
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00	n.a.	2.00
$\varepsilon(j)$	Age profile of efficiency	Ext.	Quadratic in age, PSID and CPS hump	profile	n.a.	profile
$\bar{\phi}(j)$	Baseline survival by age	Ext.	SSA period life table, ages 26 to 85	profile	n.a.	profile
p_m	Age profile of SMI prevalence	Ext.	Share with SMI, NSDUH 2023	5.5%	n.a.	5.5%
ρ_y	Persistence of log prod.	Ext.	Storesletten et al. (2004)	0.90	n.a.	0.90
σ_y	Dispersion of log prod.	Ext.	Storesletten et al. (2004)	0.20	n.a.	0.20
j_E	Last experimentation age	Ext.	Reported with and without it	55	n.a.	55
r	Risk free rate	Ext.	Kaplan, Mitman and Violante (2020)	0.03	n.a.	0.03
ξ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85	n.a.	0.85
τ_k	Capital income tax	Ext.	US long run rate on gains and dividends	0.15	n.a.	0.15
τ_ℓ	Tax level	Ext.	Clears the government budget	in eqm.	n.a.	in eqm.
G	Other public spending	Ext.	Residual at benchmark, then held fixed	in eqm.	n.a.	in eqm.
ψ	UI replacement rate	Ext.	DOL statutory replacement rates	0.50	n.a.	0.50
$\bar{y}$	Cap on insurance earnings	Ext.	DOL state benefit maxima	1.00	n.a.	1.00
κ_V	Voucher eligibility cutoff	Ext.	HUD rule	0.50	n.a.	0.50
κ_r	Tenant contribution rate	Ext.	HUD rule	0.30	n.a.	0.30
$\bar{\pi}^{ue}$	Job finding rate, annual	Ext.	Krueger, Cramer and Cho (2014)	0.55	n.a.	0.55

Note: n.a. in the Model column marks a parameter that is set from outside rather than calibrated inside the model. For those the model produces no internal target to compare with the data, so the cell reads n.a. rather than a dash, which keeps them distinct from the internally calibrated parameters.

The five calibrated inside the model. These have no source to be read off, so each is chosen to reproduce one moment, and the moment is what disciplines it.

γ, the weight on housing services. The Cobb-Douglas share of the bundle that goes on housing, so it fixes what a dwelling is worth against consumption and against the drug. Chosen to reproduce the share of renters who spend more than thirty percent of income on housing, 49.7 percent, which is the standard cost burden threshold in the 2023 American Community Survey, 21.1 of 42.5 million renter households, and is the moment a Cobb-Douglas share moves most directly. It sits here rather than in Section 3 because every life cycle model with housing has it and calibrates it the same way.

β, the discount factor. Chosen to reproduce wealth over output in the national accounts, a ratio of three. It matters more here than the description suggests, since it is what decides how heavily a death thirty years away is weighed, but it is set the way it is set everywhere. Household wealth in this model is financial wealth alone, because the rental stock sits on the rental companies' books rather than on any household's, so the counterpart in the accounts is the private capital stock rather than measured household net worth.

$\underline{h}$ **and** $\bar{h}$**, the rental ladder.** The smallest and the largest rental size. The ladder is pinned at both ends and the rung between them follows from the curvature ς, so no size is picked by hand. The two ends are matched to the two ends of the distribution of occupied units by number of bedrooms in the 2023 American Community Survey, table B25042: 13.2 percent of households are in the smallest category and 62.7 percent in the largest. Three shares carry two free numbers, so two sizes identify them and the middle share follows, which is why the curvature is set rather than calibrated.
Identifying the sizes from the size distribution rather than from the share of renters spending more than thirty or more than fifty percent of income on housing is deliberate. Those cost shares are a budget consequence of the grid rather than the grid itself, so a size calibrated on them absorbs anything else that moves the housing cost distribution: incomes, rents, the voucher, the food stamp floor. The size distribution names the object directly, and the cost shares are then available as a check. The universe is occupied units, so the model counterpart conditions on being housed: the homeless are in neither denominator.

$\bar{\pi}^{eu}$**, the job separation rate.** Chosen to reproduce the aggregate unemployment rate in the BLS Current Population Survey over 2014 to 2024, 4.85 percent, given the job finding rate set externally below. Separation is the internal one of the pair because the unemployment rate identifies the ratio and the finding rate is the better measured half.

The eighteen set from outside. Each of these is taken from a source or from statute, so a reader who disagrees is disagreeing with the literature or with the rule rather than with anything argued here. Values are in Table 6.

ς**, the curvature of the rental ladder.** Set at 1.8 rather than calibrated. With three rental sizes there are two free numbers in the size distribution, and the two ends of the ladder take them, so a third parameter chasing the same three shares would not be identified. It was tried on the largest share first and could not reach it, which is the expected result: what empties the largest unit is the level of $\bar{h}$ rather than the curvature between the ends. A uniform ladder is the restriction $\varsigma = 1$ and fits the size distribution worse.

σ**, risk aversion.** Two, from Attanasio and Weber (2010). It decides what insurance is worth, which is most of what a voucher is.

$\varepsilon(j)$**, the age profile of efficiency.** A quadratic in age fitted to the earnings hump in the PSID and the CPS, so earnings rise and then flatten rather than being constant over the life cycle.

p_m**, the age profile of SMI prevalence.** The share of adults with serious mental illness at each age, 5.5 percent overall in NSDUH 2023. The onset hazard is not read off directly but backed out of this profile and the exit rate by the identity in (22) of Section 2.7, so the profile is reproduced by construction and the level of mental illness in the model is data rather than a fitted number. It sits here rather than in Section 3 because it is the baseline profile, in the same way the survival table is: what this paper adds is ι, the multiplier a SUD puts on that onset, and that is where the interaction lives.

$\bar{\phi}(j)$**, baseline survival by age.** The SSA period life table for ages 26 to 85. This is the profile every mortality multiplier in Section 3 multiplies, so it is the base the SUD, homelessness and mental illness hazards are measured against.

ρ_y **and** σ_y**, permanent productivity.** Persistence and dispersion in logs, 0.90 and 0.20, from Storesletten et al. (2004). These decide how many households are poor enough to be at risk at all, which is why they are not inert even though they are conventional.

j_E**, the maximum age at which use can begin.** Fifty five. It is a restriction rather than an estimate, so the results are reported with and without it.

r**, the risk free rate.** Three percent, from Kaplan, Mitman and Violante (2020).

ξ**, tax progressivity.** The exponent on income in the tax schedule of Section 2.3, 0.85, from Heathcote, Storesletten and Violante (2017). It is what makes the labour income tax progressive rather than flat, and so what decides how much of either programme's bill falls on whom.

τ_k**, the capital income tax.** Fifteen percent, the long run US rate on capital gains and dividends.

τ_ℓ**, the level of the labour income tax.** Not set at all. It is whatever clears the government budget at

the benchmark and in every counterfactual, which is what makes both programmes paid for rather than free, and what makes the distortion of Section 3 a cost the model actually bears.

G**, other public spending.** The residual that balances the budget at the benchmark, then held fixed across simulations so that a policy is financed by the tax and not by cutting something else.

ψ **and** $\bar{y}$**, unemployment insurance.** The replacement rate and the cap on the earnings it is paid on, fifty percent and one times average income, both from Department of Labor statutory rates and state benefit maxima.

κ_V **and** κ_r**, the voucher rules.** The income cutoff for eligibility and the tenant's contribution, half of median income and thirty percent of income, which are HUD's own rules rather than estimates: the first is the very low income limit that governs eligibility, the second is what a tenant pays of the rent when a voucher covers the rest.

$\bar{\pi}^{ue}$**, the annual job finding rate.** 0.55, from Krueger, Cramer and Cho (2014).

3.3 Validation, and what is left undisciplined

None of the moments below is fitted. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught.

Moment	Data	Source
Homelessness rate by age, 25 to 34 through 65+	0.316 / 0.338 / 0.291 / 0.250 / 0.077 percent	HUD PIT 2024
Shares of the homeless across those bands	26.0 / 27.2 / 21.0 / 18.4 / 7.5 percent	HUD PIT 2024
Returns to homelessness within 24 months	17.4 percent	HUD SPM 2024
Drug poisonings as a share of all deaths, 25 to 44	about 20 percent	CDC WONDER
Bottom half's share of net worth	2.52 percent	Fed DFA
Housing costs above 30 and above 50 percent of income	40.6 and 17.5 percent	SCF 2022
Age weighted standardised mortality ratio, SUD	2.5 to 9.8	Degenhardt et al.
Age weighted standardised mortality ratio, homeless	9.1 / 4.5 / 1.1 by band	Baggett et al.
Tenant contribution rate	31.2 percent	HUD
Voucher holders with wages as their main income	28 percent	HUD
Income with a moderate or severe SUD, over average	71.1 percent	NSDUH 2024
Labour force participation, SUD against clean	65.8 against 61.5 percent	NSDUH 2024
Homeless at treatment admission	14.5 percent	TEDS-A 2022
Co-occurring psychiatric problem at admission	43.1 percent	TEDS-A 2022
Working at all, among the homeless in treatment	6.6 percent	TEDS-A 2022
Drug use disorder share of the homeless	21.7 percent	Gutwinski et al. (2021)
Drug price participation elasticity	-0.63	Saffer and Chaloupka (1999)

Three parameters remain undisciplined, and they are flagged rather than hidden: the treatment effect on recovery g, the recovery multiplier for SMI μ_m, and the drug supply curvature ν_2. Of these only g carries a headline result, and it is reported as a range rather than as a number.

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